Aakash Shahani

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Education

University of South Florida (USF) Graduation: December 2025
Bachelor of Science in Computer Science *GPA: 3.5/4.0*

- Relevant coursework: Data Structures & Algorithms, Operating Systems, Distributed Systems, Database Design, Big Data Storage & Analysis, Probability & Statistics, Software Engineering

Technical Skills

Languages: Python, Go, C/C++, Java, TypeScript, JavaScript, SQL, Bash
AI / ML: PyTorch, transformers, Hugging Face, scikit-learn, XGBoost, RAG, Embeddings, LangChain, Prompt Engineering, LLM APIs (Anthropic/OpenAI), SHAP
Data & Cloud: Airflow, dbt, Iceberg, Spark, Kafka, star schema & SCD2, Great Expectations, AWS (S3/Glue/Athena), Terraform, Docker, CI/CD (GitHub Actions), pandas, NumPy, A/B Testing, Prometheus/Grafana
Backend, Databases & Tools: FastAPI, REST, GraphQL, Node.js, Next.js, React, PostgreSQL, MySQL, MongoDB, Redis, pgvector/FAISS, Metabase, Git, Linux, pytest

Experience

AI/ML Engineer Jan 2026 – Present
CSSAI Lab, University of South Florida *Tampa, FL*

- Built an end-to-end LLM evaluation pipeline (temperature-0 batch inference, versioned datasets, automated scoring) benchmarking frontier models (GPT-4.1, Gemini 2.5 Flash, Claude Sonnet 4.6) against fine-tuned transformer and classical baselines in one reproducible run.
- Drove the detector to **0.79** macro-F1 / **0.81** accuracy, significantly beating every trained baseline (XGBoost, fine-tuned DistilBERT) and approaching human-expert agreement, through precision-first prompt engineering, grouped cross-validation, and bootstrapped significance testing.

Machine Learning Research Intern Aug 2024 – Apr 2025
University of South Florida *Tampa, FL*

- Engineered an end-to-end ETL pipeline pulling protein sequences from the UniProt REST API and applying MMseqs2 clustering at a **30%** similarity threshold to deduplicate and standardize a model-ready dataset.
- Designed and trained an LSTM binding-site classifier (BCE loss, Adam with grid search, dropout, LR scheduling), benchmarking against CLAPE-SMB and SCRIBER with k-fold cross-validation and a held-out test set.

Projects

ScholarLens: Agentic Research Platform | *FastAPI, Next.js, pgvector, Claude API* GitHub | Live Demo

- Built an agentic RAG platform that flags cross-paper contradictions via a two-stage BM25 + dense pre-filter and an LLM judge at macro-F1 **0.788** / κ **0.683**, cutting ~**4,500** LLM calls by over **90%**.
- Ran **6** analyses per paper concurrently for a **5**× speedup and served pgvector (HNSW) search behind a cross-encoder reranker at nDCG@5 **0.98**; deployed live with multi-user isolation.

TradePulse: Real-Time Streaming Pipeline | *Kafka, Spark Streaming, Docker* GitHub

- Ingested live Coinbase trades through a **6**-partition Kafka topic into 1-minute OHLC candles in Spark Structured Streaming; guaranteed exactly-once output via watermarks and idempotent writes, proven by a mid-stream crash test with **0** duplicate candles.
- Sustained sub-second **600–800ms** micro-batches under ~**1,500** rows/sec bursts across **9** Dockerized services with Prometheus/Grafana observability; validated by **20** tests in CI.

NewsRec: Cold-Start News Recommender | *PyTorch, FAISS, FastAPI, Docker* GitHub

- Trained a content-based two-tower retriever on the MIND dataset that beat popularity/SVD/ALS baselines by **+0.124** AUC (~**23%**) across **73,152** impressions by embedding article text to solve cold-start.
- Served candidate generation at **2.8ms** p99 over **3,000** users via a FAISS HNSW index so the FastAPI path never loads a model at request time; Docker + **17** tests in CI.

UrbanFlow: Data Lakehouse Platform | *Airflow, dbt, Apache Iceberg, AWS, Terraform* GitHub

- Built an end-to-end data lakehouse over **3M** NYC taxi trips: Airflow into Apache Iceberg on S3, dbt-modeled into a star schema (fact + SCD Type 2 dims), deployed on AWS with Terraform and queried on Athena/Trino.
- Cut Athena bytes scanned **97%** (**4.81**→**0.13**MB) via day-partitioning; gated it with Great Expectations and **17** dbt tests that block bad data before the gold layer; idempotent Iceberg MERGE held **2.87M** rows at **0** duplicates.